	B.M.S. COLLEGE OF ENGINEERING, BANGALORE-19 Computer Science & Engineering	
INTERNALS-1		
Course Code: 23CS3PCDST	**Course Title: Data Structures**	
Semester: 3	**Maximum Marks: 40**	**Date: 6-11-2024**
Faculty Handling the Course:	**Dr. BSR, Dr. SKS, NM, RAD, LN, GN, Dr. JSN**	
Instructions: Internal choice provided in Part C.		

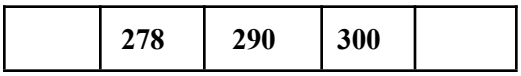
PART-A

Total 5 Marks (No Choice)

Q.No.	Question	Marks	CO-PO
1.	Write an algorithm to evaluate the postfix expression using stack data structure.	5	1-1

PART-B

Total 15 Marks (No Choice)

Q.No.	Question	Marks	CO-PO
2 a)	Consider the elements are inserted using circular queue. Here front is at index 1 and rear is at index 3  The deleted element of the queue is pushed into the stack. Currently the stack is having one element 266. Perform the following operation and pictorially represent the status of queue and stack after every operation: insert(310) insert(400) delete() insert(500) delete()	**5**	**2-2**
2 b)	Convert the following infix expression to postfix expression and show all the steps clearly. **A+B*(C^D-E)/F**	**5**	**1-1**

2 c)	Analyze the below given C code. Write the complete stack trace and output. (n = 4721)
	
	int func(int n)
	{
	if (n == 0)
	{
	return 0;
	}
	return (n % 10) + func(n / 10);
	}
		5	**2-2**

PART- C

Total 20 Marks

Q. No.	Question	Marks	CO-PO
3 a)	Design a C Program using suitable data structure to print the Next Greater Element for every element in a given array. The Next Greater Element for an element x is the first greater element on the right side of x in the array. Elements for which no greater element exist, consider the next greater element as -1. Sample input: {11, 13, 21, 3} Output:{13, 21, -1, -1}	**10**	**3-3**
	OR		
3 b)	Design a C program using suitable data structure to find first non-repeating character each time a character is inserted to the string. If character is repeated, we output as -1 else print the non-repeating character. Sample Input: a a b c Output: a -1 b c	**10**	**3-3**
4a)	Imagine a system that has to process multiple tasks, each with a priority level. The task with the **lowest priority number** should be processed first. Once a task is completed, it should be removed from the queue, and the next task with the lowest priority should be processed. Develop an application (C Program) for the above scenario.	**10**	**3-3**
	OR		
4b)	A shop manages its chocolate inventory. Each box of chocolates contains a positive number of chocolates. Initially, the inventory is empty. Over the next N minutes, the following two operations may occur: **1.** Sell Operation: The box of chocolates on top of the inventory gets sold. **2.** Receive Operation: A new box of chocolates is received from the warehouse and placed on top of the inventory. For each Sell Operation, output the number of chocolates sold (the number of chocolates in the box). If a sell operation is attempted on an empty inventory, output "No chocolates available". For each Receive Operation, if the inventory reaches its maximum capacity (10 boxes), output "Inventory Full for the Day" and do not add any more boxes. Develop C code to simulate the inventory management system using suitable data structure.	**10**	**3-3**

*****ALL THE BEST*****